



AZ-104

Administer Monitoring

AZ-104 Agenda

- 01: Administer Identity
- 02: Administer Governance and Compliance
- 03: Administer Azure Resources
- 04: Administer Virtual Networking
- 05: Administer Intersite Connectivity
- 06: Administer Network Traffic Management
- 07: Administer Azure Storage
- 08: Administer Azure Virtual Machines
- 09: Administer PaaS Compute Options
- 10: Administer Data Protection
- 11: Administer Monitoring

Learning Objectives - Administer Monitoring

- Introduction to Azure Monitor
- Improve incident response with alerting on Azure
- Analyze your infrastructure using Azure Monitor logs
- Lab 11 – Implement Monitoring

Introduction to Azure Monitor



Learning Objective – Introduction to Azure Monitor

- Describe Azure Monitor Key Capabilities
- Describe Azure Monitor Components
- Define Metrics and Logs
- Identify Data Types
- Describe Activity Log Events
- Learning Recap

Monitor resources in Azure

- Interpret metrics in Azure Monitor
- Configure log settings in Azure Monitor
- Configure and interpret monitoring of virtual machines, storage accounts, and networks by using Azure Monitor Insights

Describe Azure Monitor Key Capabilities



Monitor & Visualize Metrics

Metrics are numerical values available from Azure Resources helping you understand the health, operation & performance of your systems.

[Explore Metrics](#)

Core monitoring for Azure services



Query & Analyze Logs

Logs are activity logs, diagnostic logs and telemetry from monitoring solutions; Analytics queries help with troubleshooting & visualizations.

[Search Logs](#)

Collects metrics, activity logs, and diagnostic logs



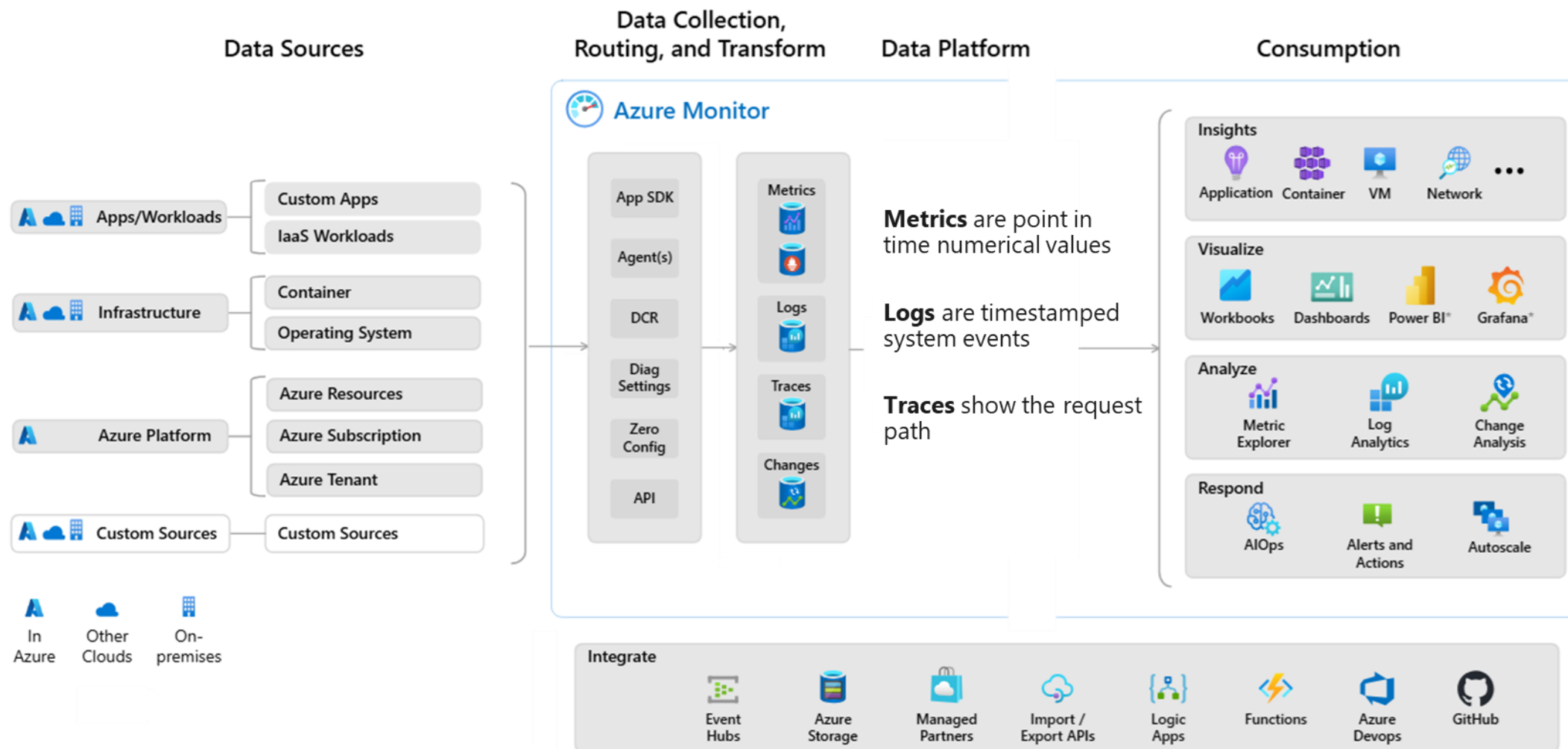
Setup Alert & Actions

Alerts notify you of critical conditions and potentially take corrective automated actions based on triggers from metrics or logs.

[Create Alert](#)

Use for time critical alerts and notifications

Understand Azure Monitor Components



Define Metrics and Logs



- Metrics are numerical values that describe some aspect of a system at a point in time
- They are lightweight and capable of supporting near real-time scenarios



- Logs contain different kinds of data organized into records with different sets of properties for each type
- Telemetry (events, traces) and performance data can be combined for analysis

Describe Activity Log Events

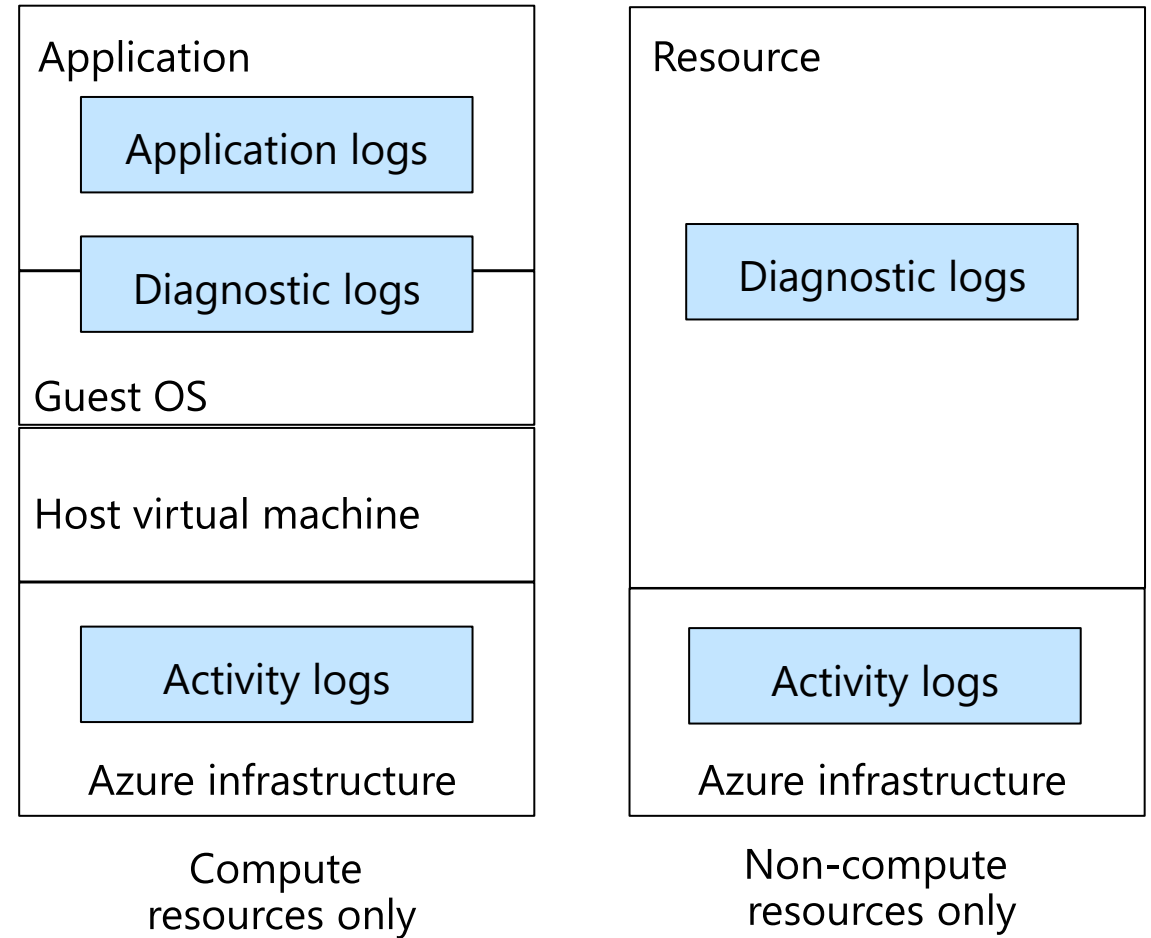
Send data to Log Analytics for advanced search and alerts

Query or manage events in the Portal, PowerShell, CLI, and REST API

Stream information to Event Hub

Archive data to a storage account

Analyze data with Power BI



Query the Activity Log

Activity log

 Edit columns

 Refresh

 Diagnostics settings

 Download as CSV

 Logs

 Pin current filters

 Search

 Quick Insights

 Add Filter

Management Group : **None**

Subscription : **2 selected**

Timespan : **Last 6 hours**

Event severity : **All**

Operation name	Status	Time	Time stamp	Subscription
>  Create or Update Virtual Network Subnet	Failed	a minute ago	Thu Mar 12 ...	ASC DEMO
>  Write GuestConfigurationAssignments	Succeeded	17 minutes ...	Thu Mar 12 ...	ASC DEMO
>  Gets workflow recommend operation groups	Succeeded	29 minutes ...	Thu Mar 12 ...	ASC DEMO

Filter by Management group, Subscription, Timespan, and Event Severity

Add a filter, like Event Category (Security, Recommendations, Alerts)

Pin current filters and download as CSV

Improve incident response with alerting on Azure



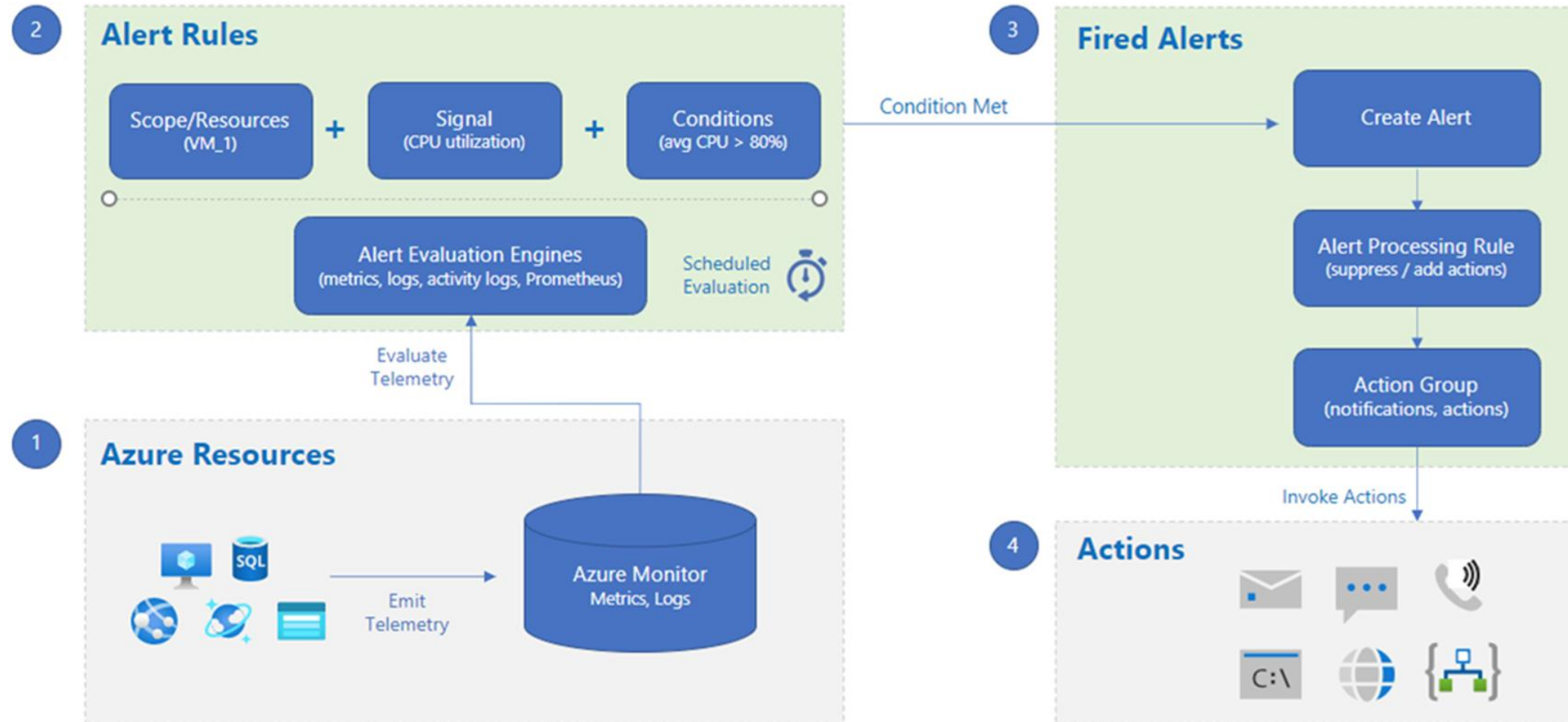
Learning Objective - Improve incident response with alerting

- Manage Azure Monitor Alerts
- Create Alert Rules
- Create Action Groups
- Demonstration – Alerts
- Learning Recap

Monitor resources in Azure

- Set up alert rules, action groups, and alert processing rules in Azure Monitor

Manage Azure Monitor Alerts



Create Alert Rules

[Home](#) > [Monitor | Alerts](#) >

Alert rules ...

Name ↑↓	Condition	Severity ↑↓	Target scope	Target resource type	Signal type ↑↓	Status ↑↓
☐ AzureSecurityCenter	Table rows > 1	4 - Verbose	export2LogA	Log Analytics workspace	Log search	✔ Enabled
☐ CPU Usage Percentage	node_cpu_usage_percentage > 80	3 - Informational	Demo	Kubernetes service	Metrics	✔ Enabled
☐ Failure Anomalies - HumanResources	Failure Anomalies detected	3 - Informational	humanresources	Application Insights	Smart detector	✔ Enabled

- Alert rules combine the resources to be monitored, the signal or data from the resource, and the conditions.
- You can enable recommended out-of-the-box alert rules in the Azure portal.

Create Action Groups

Defines a set of notifications and/or actions when an alert is triggered

You can add up to five action groups to an alert rule. Multiple alert rules can use the same action group.

Notifications

Configure the method in which users will be notified when the action group triggers. Select notification types, provide receiver details and add a unique description. This step is optional.

Notification type ⓘ	Name ⓘ	Selected ⓘ
---------------------	--------	------------

Email Azure Resource Manager Role		
Email/SMS message/Push/Voice		

Actions

Configure the method in which actions are performed when the action group triggers. Select action types, fill out associated details, and add a unique description. This step is optional.

Action type ⓘ	Name ⓘ	Selected ⓘ
---------------	--------	------------

Automation Runbook		
Azure Function		
Event Hub		
ITSM		
Logic App		
Secure Webhook		
Webhook		

Demonstration – Alerts

- Create and configure an alert rule
- Review alerts

The screenshot shows the Azure Monitor Alerts interface. On the left sidebar, the 'Monitor' link is highlighted with a red box and a red circle containing the number 1. In the main navigation pane, the 'Alerts' option is highlighted with a red box and a red circle containing the number 2. In the top right corner, the 'Create' button is highlighted with a red box and a red circle containing the number 3, with a dropdown menu showing 'Alert rule', 'Action group', and 'Alert processing rule'.

Home > Monitor

Monitor | Alerts

Microsoft

Search (Ctrl+/)

Overview
Activity log
Alerts
Metrics
Logs
Service Health
Workbooks

Insights

- Applications
- Virtual Machines
- Storage accounts
- Containers
- Networks
- SQL (preview)

+ Create

- Alert rule**
- Action group
- Alert processing rule

Alert rules

Error Warning

273 67 199 0

Name ↑↓	Severity ↑↓
☐ High CPU alert monito...	1 - Error
☐ High CPU alert monito...	1 - Error
☐ Alert for all VMs in sel...	1 - Error
☐ Alert for all VMs in sel...	1 - Error
☐ Contoso Customer Ch...	1 - Error
☐ VMHealth monitor 'ro...	1 - Error
☐ Contoso Retail VM Em...	1 - Error
☐ Computers with high	0 - Critical



Analyze your infrastructure using Azure Monitor logs



Learning Objective - Analyze your infrastructure using Azure Monitor logs

- Determine Log Analytics Uses
- Create a Workspace
- Query Log Analytics Data
- Structure Log Analytics Queries
- Demonstration – Log Analytics
- Learning Recap

Monitor and maintain Azure resources (10–15%): Monitor resources in Azure

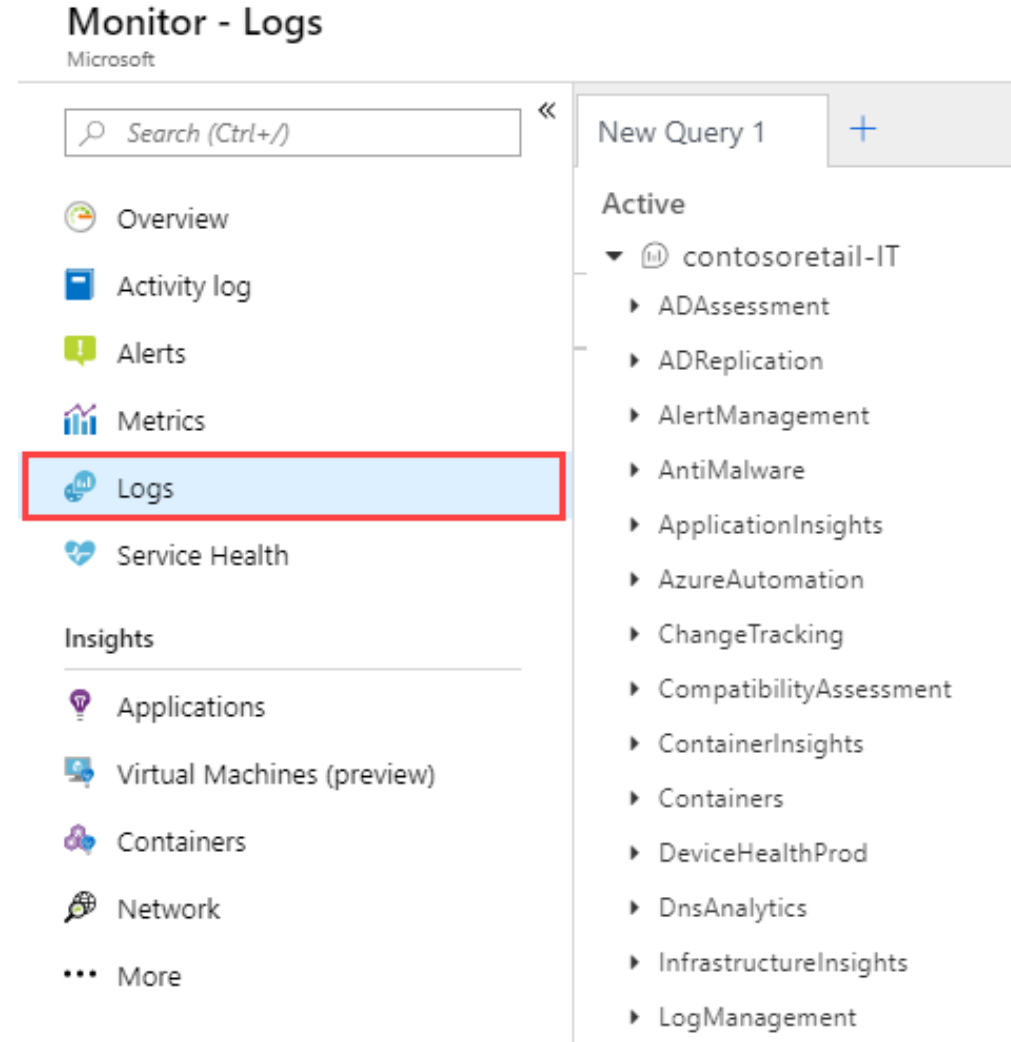
- Query and analyze logs in Azure Monitor

Determine Log Analytics Uses

A service that helps you collect and analyze data generated by resources in your cloud and on-premises environments

Write log queries and interactively analyze their results

Examples include assessing system updates and troubleshooting operational incidents



Create a Workspace

A workspace is an Azure resource and is a container where data is collected, aggregated, analyzed, and presented

You can have multiple workspaces per Azure subscription, and you can have access to more than one workspace

A workspace provides a geographic location, data isolation, and scope

[Home](#) > [Log Analytics workspaces](#) >

Create Log Analytics workspace ...

Basics Tags Review + Create

i A Log Analytics workspace is the basic management unit of Azure Monitor Logs. There are specific considerations you should take when creating a new Log Analytics workspace. ×

With Azure Monitor Logs you can easily store, retain, and query data collected from your monitored resources in Azure and other environments for valuable insights. A Log Analytics workspace is the logical storage unit where your log data is collected and stored.

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * **i**

ASC DEMO ▼

Resource group * **i**

▼

[Create new](#)

Instance details

Name * **i**

Region * **i**

East US 2 ▼

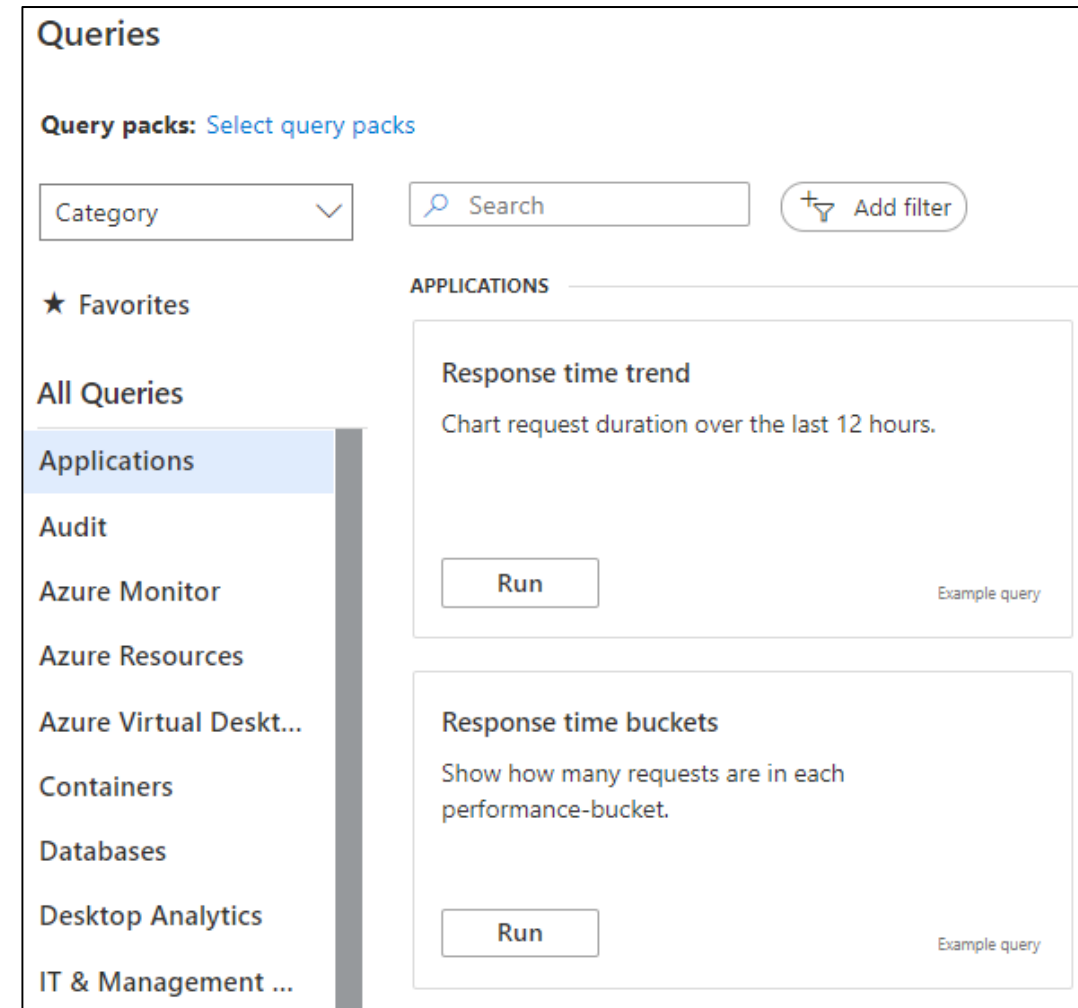
Query Log Analytics Data

Common queries and a query language (KQL) for custom searches

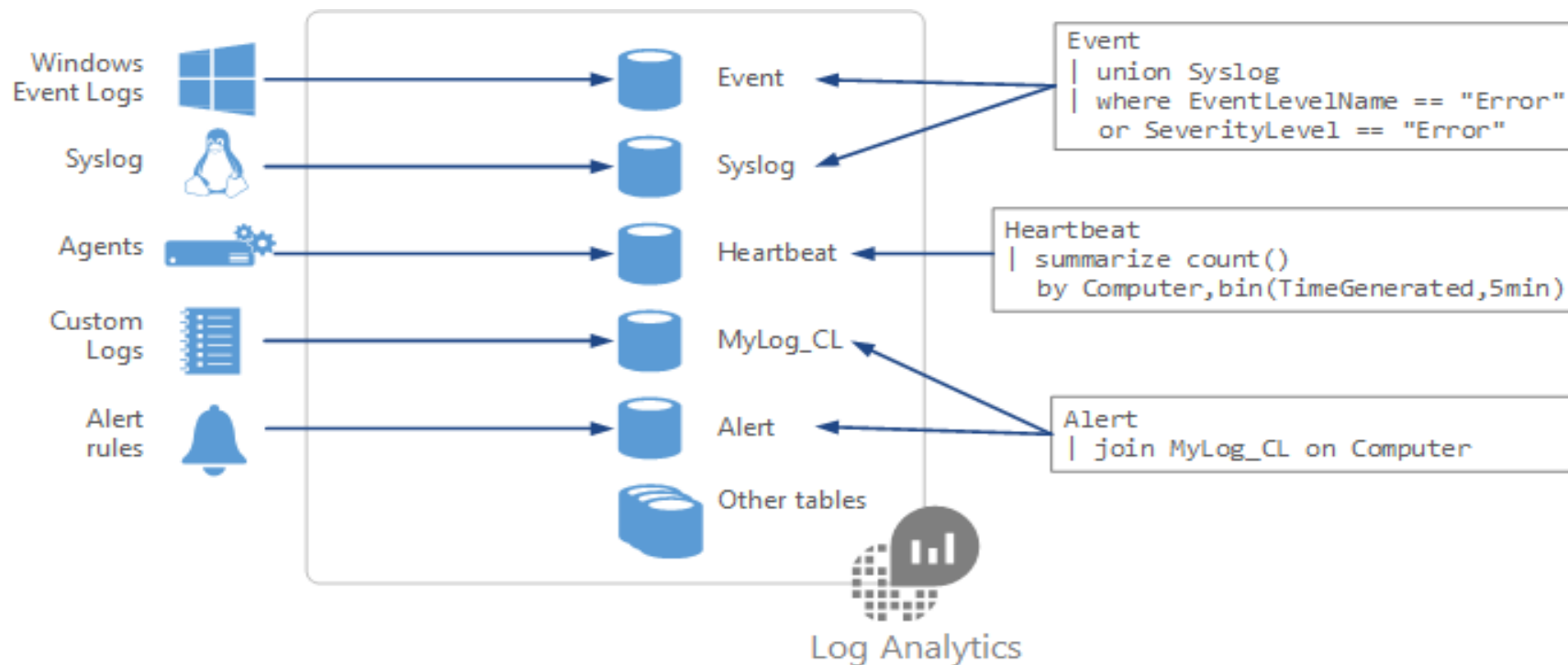
Quickly retrieve and consolidate data in the repository

Save or have log searches run automatically to create an alert

Export the data to Power BI or Excel



Structure Log Analytics Queries

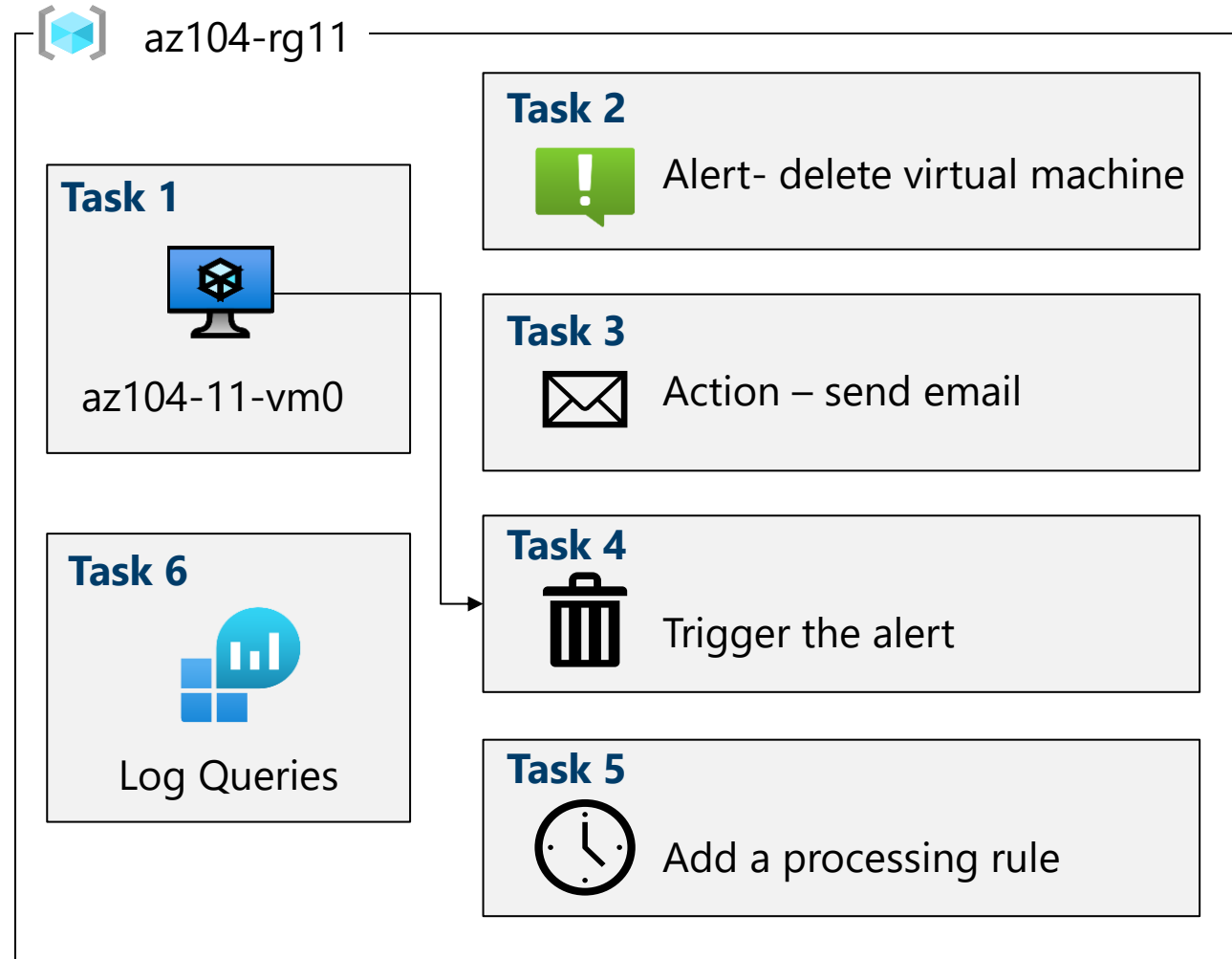


```
Event  
| where (EventLevelName == "Error")  
| where (TimeGenerated > ago(1days))  
| summarize ErrorCount = count() by Computer  
| top 10 by ErrorCount desc
```

Lab – Implement Monitoring



Lab 11 – Architecture diagram



End of presentation



